Software Requirements Specification (SRS) Audit

Use Case Model vs. Implementation Audit & Update Review

Course: NTU SC2006 Software Engineering
Project: Smart Parking Recommendation App (Team 1)
SRS Version: 0.3 → 0.4 Proposal
Audit Date: October 2026

1. Executive Summary & Verdict

This audit evaluates the current **Appendix B.1 Use Case Diagram** and **B.2 Use Case Descriptions** (Version 0.3) against the concrete codebase implementation across `apps/backend`, `apps/frontend`, and `packages/shared-types`.

Area Audited	Status	Key Findings & Recommended Actions
Core Functional Scope	FULLY ALIGNED	All 16 use cases (`UC-01` to `UC-16`) have concrete backend routes and frontend UI implementations. The functional scope accurately reflects the built product.
Actor Generalization	FULLY ALIGNED	The actor inheritance `Registered Driver –▷ Driver` perfectly matches the codebase's anonymous/guest mode (`session.isGuest = true`) vs. authenticated sessions.
Secondary System Actors	NEEDS UPDATE	**Supabase Auth** is missing from the diagram despite powering `UC-12 (Register/Login)` and `UC-13 (Manage Profile)`. Browser Geolocation should be explicitly bound to `UC-01` & `UC-03`.
Filtering Relationship (UC-07)	NEEDS UPDATE	`UC-07 (Filter Carparks)` is currently drawn as a detached use case. In UML and code, filtering is an `<<extend>>` to `UC-01 (Search Destination)`.
Weather Dependencies (UC-05)	NEEDS UPDATE	`UC-05 (Get Weather Forecast)` is currently only included by `UC-15`. However, the codebase and UI display live weather warnings in both `UC-09 (Navigation)` and `UC-06 (Carpark Details)`.
Alert Lifecycle (UC-14)	NEEDS UPDATE	Labelled passively as "*Receive Availability Alert*". In reality, the user configures/subscribes to monitored carparks in `CarparkDetailPage`, and receives toast alerts in the app shell.

> **Final Verdict: Highly Viable, Requires UML Precision Tuning**
> The existing SRS use case foundation does **not** require a conceptual rewrite. However, updating the diagram and descriptions to correct UML association semantics (specifically `<<extend>>`, missing secondary actors, and multi-consumer weather/routing inclusions) will eliminate grading penalties under NTU SC2006 OOAD criteria.

2. Detailed Subsystem Analysis: SRS vs. Codebase

2.1 User & Authentication Subsystem (UC-12, UC-13, UC-10, UC-11)

- **Guest vs. Registered Actor:** In `AuthContext.tsx`, unauthenticated drivers can perform destination searches, view carparks, apply filters, and navigate. `Registered Driver` inherits these and gains access to `/api/favourites` (UC-10, UC-11), `/api/users/me` (UC-13), and `/api/alerts` (UC-14). This cleanly verifies the actor inheritance in UML.
- **Missing External Identity Provider:** `UC-12` and `UC-13` connect only to the human actor. In production, requests hit Supabase GoTrue Auth (`apps/backend/src/modules/auth/auth.service.ts`). An external actor `<<system>> Authentication Service (Supabase Auth)` must be added.

- **Favourites to Search Flow:** In `FavouritesPage.tsx`, tapping a saved location directly triggers a search for that destination. In UML, this can be represented as `UC-11 •••> UC-01 <<invokes>>`.

2.2 Destination Search & Recommendation Subsystem (UC-01, UC-07, UC-15, UC-16)

- **Search Orchestration (UC-01 & UC-15):** `search.service.ts` acts as the orchestrator endpoint (`GET /api/carparks/nearby`). When `sort = 'recommended'`, it delegates to `rankCarparks()` (UC-15). UC-01 rightly includes UC-15.
- **Filtering Semantics (UC-07):** The SRS lists UC-07 as a standalone use case. However, carparks cannot be filtered without candidate carparks being retrieved first. UC-01 alternative flow `UC-01.AC.2` explicitly notes "*Driver applies filters*". Therefore, `UC-07` must extend `UC-01` via `<<extend>>`.
- **Historical Availability (UC-16):** Matches the code in `historical-availability.ts`, validating that at least 4 distinct historical records exist for the same day-of-week and hour bucket before returning an availability ratio.

2.3 Route & Navigation Subsystem (UC-03, UC-04, UC-08, UC-09)

- **Multi-Modal Route Composition:** In `NavigationPage.tsx` and `useNavigationRoutes.ts`, navigation comprises both a driving leg ($Origin \rightarrow Carpark$, UC-03) and a walking leg ($Carpark \rightarrow Destination$, UC-04). UC-09 correctly includes UC-03 and UC-04.
- **Live Weather in Navigation:** `NavigationPage.tsx` imports `useWeather` to warn drivers: "*Showers expected... Bring an umbrella*" (SRS page 15). The current diagram lacks this connection. `UC-09` must include `UC-05 (Get Weather Forecast)`.
- **Carpark Selection (UC-08):** In `CarparkCard.tsx`, tapping "Navigate" transitions directly to navigation (UC-09), while tapping the card opens details (UC-06). UC-08 is simply an interaction step that should bridge into UC-06 and UC-09.

2.4 Carpark Availability & Alerts Subsystem (UC-02, UC-06, UC-14)

- **Real-Time Data Poller:** `UC-02 (Get Real-Time Carpark Information)` is fed by background polling every 60 seconds (`availability/poller.ts`). It is correctly included by `UC-15 (Recommendations)` and `UC-06 (Details)`.
- **Alert Management (UC-14):** In `CarparkDetailPage.tsx`, registered users toggle availability notifications via `AvailabilityAlertControl`. The backend checks threshold tier transitions (`hasSignificantAvailabilityChange`). Naming should be updated to "*Manage & Receive Availability Alerts*".

3. Granular Use Case Cross-Reference Matrix

ID	Use Case Name	Associated Actors	Diagram Dependencies	Codebase Implementation
UC-01	Search Destination & View Recommended Carparks	Driver, Geolocation, Routing/Places Service	`<<include>> UC-15` `<<extend>> UC-07 (Suggested)`	`SearchPage.tsx`, `search.routes.ts`, `destinations.service.ts`
UC-02	Get Real-Time Carpark Information	Carpark Data Service (data.gov.sg / LTA)	Included by UC-15, UC-06, UC-14	`availability/service.ts`, `availability/poller.ts`
UC-03	Calculate Driving Time	Routing / Map Service (Google Routes)	Included by UC-15, UC-09	`routing.service.ts` (`getDrivingRoute`)
UC-04	Calculate Walking Distance & Time	Routing / Map Service (Google Routes)	Included by UC-15, UC-09	`routing.service.ts` (`getWalkingRoute`)
UC-05	Get Weather Forecast	Weather Service (data.gov.sg NEA)	Included by UC-15, **UC-09**, **UC-06**	`weather.service.ts`, `useWeather.ts`
UC-06	View Carpark Details	Driver	`<<include>> UC-02`, `<<include>> UC-05`	`CarparkDetailPage.tsx`, `/api/carparks/:carParkNo`
UC-07	Filter Carparks	Driver	**`<<extend>> UC-01`**	`CarparkFilters.tsx`, `search.service.ts` (passesFilters)
UC-08	Select Carpark	Driver	Bridges into UC-06 & UC-09	`CarparkCard.tsx` (card select / navigation trigger)
UC-09	View Route to Carpark & Destination	Driver, Routing/Map Service	`<<include>> UC-03, UC-04, UC-05`	`NavigationPage.tsx`, `useNavigationRoutes.ts`
UC-10	Save Favourite Location	Registered Driver	`<<extend>> UC-06 / UC-01`	`favouritesApi.ts`, `POST /api/favourites`
UC-11	Manage Saved Locations	Registered Driver	`<<invokes>> UC-01`	`FavouritesPage.tsx`, `useFavouriteDeletion.ts`
UC-12	Register / Login	Driver, **Supabase Auth**	Standalone entry point	`SignInPage.tsx`, `SignUpPage.tsx`, `auth.service.ts`
UC-13	Manage User Profile	Registered Driver, **Supabase Auth**	Standalone user setting	`ProfilePage.tsx`, `PUT/DELETE /api/users/me`
UC-14	Manage & Receive Availability Alerts	Registered Driver	`<<include>> UC-02`, `<<extend>> UC-06`	`alerts.service.ts`, `AvailabilityAlertsProvider.tsx`
UC-15	Generate Carpark Recommendations	Internal Orchestrator	`<<include>> UC-02, 03, 04, 05, 16`	`rank-carparks.ts`, `carpark-ranking.context.ts`
UC-16	Get Historical Carpark Availability	Carpark Data Service	Included by UC-15	`historical-availability.ts`

4. Actionable Update Checklist for SRS Version 0.4

1. **Update Figure B.1 (Use Case Diagram):**
   - Add secondary actor `<<system>> Authentication Service (Supabase Auth)` linked to `UC-12` and `UC-13`.
   - Modify relationship from `UC-07` to `UC-01`: make it a dashed arrow pointing to `UC-01` labelled `<<extend>>`.
   - Add an `<<include>>` dashed arrow from `UC-09` to `UC-05 (Get Weather Forecast)`.
   - Add an `<<include>>` dashed arrow from `UC-06` to `UC-05 (Get Weather Forecast)`.
   - Ensure `Routing & Places Service` is connected to `UC-01` for Google Places destination search/resolution.
2. **Update B.2 Use Case Descriptions:**
   - **UC-09 (View Route to Carpark and Destination):** Update the *Includes* field to: `UC-03 Calculate Driving Time, UC-04 Calculate Walking Distance and Time, UC-05 Get Weather Forecast`.
   - **UC-06 (View Carpark Details):** Under *Includes*, add `UC-05 Get Weather Forecast` to align with the weather display shown in prototype screenshot page 13.
   - **UC-14 (Receive Availability Alert):** Rename to "*Manage & Receive Availability Alert*" and describe the initial setup step: "*The driver toggles availability monitoring on the carpark details view*".
3. **Note on Design Patterns in B.4 / B.5:**
   - In the **Design Class Model (B.4)** and **Architecture (B.5)**, explicitly document how `UC-15` uses the **Strategy Pattern** (`BalancedRankingStrategy`, `ProximityFirstStrategy`, `MaxAvailabilityStrategy`) and how routing uses the **Factory Method**.

5. Updated PlantUML Source Code for B.1

Copy and paste the following PlantUML script into PlantText, PlantUML Server, or your local VS Code PlantUML extension to generate the updated diagram:

```
@startuml
skinparam packageStyle rectangle
skinparam shadowing false
skinparam roundcorner 12
skinparam monochrome false

' SC2006 Theme Palette
skinparam ActorBorderColor #1e3a8a
skinparam ActorBackgroundColor #dbeafe
skinparam ActorFontColor #0f172a
skinparam ActorFontStyle bold

skinparam usecase {
  BackgroundColor #ffffff
  BorderColor #2563eb
  FontColor #0f172a
  FontSize 11
  ArrowColor #475569
  ArrowFontSize 10
}

left to right direction

' ========================================
' ACTORS
' ========================================
actor "Driver\n(Guest User)" as Driver
actor "Registered Driver" as RegDriver

actor "Authentication Service\n(Supabase Auth)" as AuthService <>
actor "Carpark Data Service\n(data.gov.sg / LTA)" as CarparkService <>
actor "Weather Service\n(data.gov.sg NEA)" as WeatherService <>
actor "Routing & Places Service\n(Google Maps Platform)" as RoutingService <>
actor "Browser Geolocation API" as GeoService <>

' Actor Inheritance
RegDriver -up-|> Driver

' ========================================
' SYSTEM BOUNDARY
' ========================================
rectangle "Smart Parking Recommendation Application" {

  ' Authentication & Profile
  usecase "UC-12: Register / Login" as UC12
  usecase "UC-13: Manage User Profile" as UC13

  ' Favourites
  usecase "UC-10: Save Favourite Location" as UC10
  usecase "UC-11: Manage Saved Locations" as UC11

  ' Primary Search & Filtering
  usecase "UC-01: Search Destination and\nView Recommended Carparks" as UC01
  usecase "UC-07: Filter Carparks" as UC07

  ' Carpark Details & Selection
  usecase "UC-06: View Carpark Details" as UC06
  usecase "UC-08: Select Carpark" as UC08
  usecase "UC-14: Manage & Receive\nAvailability Alerts" as UC14

  ' Navigation & Routes
  usecase "UC-09: View Route to Carpark\nand Destination" as UC09

  ' Subordinate & Algorithmic Use Cases
```

```
  usecase "UC-15: Generate Carpark\nRecommendations" as UC15
  usecase "UC-02: Get Real-Time\nCarpark Information" as UC02
  usecase "UC-03: Calculate Driving Time" as UC03
  usecase "UC-04: Calculate Walking\nDistance & Time" as UC04
  usecase "UC-05: Get Weather Forecast" as UC05
  usecase "UC-16: Get Historical\nCarpark Availability" as UC16
}

' ========================================
' ACTOR ASSOCIATIONS
' ========================================
Driver --> UC12
Driver --> UC01
Driver --> UC06
Driver --> UC08
Driver --> UC09

RegDriver --> UC13
RegDriver --> UC10
RegDriver --> UC11
RegDriver --> UC14

' External System Associations
UC12 --- AuthService
UC13 --- AuthService

UC01 --- GeoService
UC01 --- RoutingService

UC02 --- CarparkService
UC16 --- CarparkService

UC05 --- WeatherService

UC03 --- RoutingService
UC04 --- RoutingService
UC09 --- RoutingService

' ========================================
' USE CASE DEPENDENCIES (<> & <>)
' ========================================
UC07 .> UC01 : <>
UC11 .> UC01 : <>

UC01 .> UC15 : <>

UC15 .> UC02 : <>
UC15 .> UC03 : <>
UC15 .> UC04 : <>
UC15 .> UC05 : <>
UC15 .> UC16 : <>

UC06 .> UC02 : <>
UC06 .> UC05 : <>

UC10 .> UC06 : <>
UC14 .> UC06 : <>
UC14 .> UC02 : <>

UC09 .> UC03 : <>
UC09 .> UC04 : <>
UC09 .> UC05 : <>

@enduml
```